

COURSE DESCRIPTION FORM

INSTITUTION FAST School of Computing, National University of Computer and Emerging Sciences, Islamabad

PROGRAM(S) TO BE EVALUATED BS-CS Spring 2024

Course Description

(Fill out the following table for each course in your computer science curriculum. A filled-out form should not be more than 2-3 pages.)

Course Code	CS-1004															
Course Title	Object-Oriented Programming (in C++)															
Credit Hours	3															
Prerequisites by Course(s) and Topics	Programming Fundamentals (CS-1002)															
Grading Policy	Absolute grading scheme															
Policy about missed assessment items in the course	Retake of missed assessment items (other than midterm/ final exam) will not be held. For a missed midterm/ final exam, an exam retake/ pretake application along with necessary evidence are required to be submitted to the department secretary. The examination assessment and retake committee decide the exam retake/ pretake cases.															
Policy about late submission of assessment item	Penalties for late submissions of assignment/project: 1. Up to 15 mins, loss of 20% of the mark awarded. 2. After 15 mins, assignment/project will not be accepted for marking.															
Course Plagiarism Policy	Plagiarism in midterm or final exams may lead to an automatic grade of 'F' for the entire course. Plagiarism in any assignment item (including assignments and quizzes) will result in a score of zero for that item category. Plagiarism in the project component will result in a score of zero for all assignment items, including assignments, quizzes, and the project itself. If the aforementioned violation occur multiple times, the instructor reserves the right to refer the case to the Department Disciplinary Committee, where the minimum penalty may be the award of an 'F' grade for the course.															
Assessment Instruments with Weights (homework, quizzes, midterms, final,	<table border="1"> <tr> <th colspan="3">Assessment items of Theory Part</th> </tr> <tr> <th>Assessment Item</th><th>Number</th><th>Weight (%)</th> </tr> <tr> <td>Quizzes</td><td>>=5</td><td>14</td> </tr> <tr> <td>Assignments</td><td>>=3</td><td>06</td> </tr> <tr> <td>Project</td><td>1</td><td>10</td> </tr> </table>	Assessment items of Theory Part			Assessment Item	Number	Weight (%)	Quizzes	>=5	14	Assignments	>=3	06	Project	1	10
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programming assignments, lab work, etc.)	Sessional – I	1	15																					
	Sessional – II	1	15																					
	Final Exam	1	40																					
	Grading Policy: Absolute grading policy will be used for this course.																							
Course Instructors	Dr. Ali Zeeshan Ijaz, Miss Mariam Hida, Mr. Shehreyar Rashid,																							
Course Coordinator	Mr. Shehreyar Rashid																							
URL (if any)	Google Classroom Link: https://classroom.google.com/c/NjU0NDgxMjUxOTkx?cjc=aeu7rrz Google Classroom Code: aeu7rrz																							
Current Catalog Description	The course aims to provide students with the ability to analyze the given requirements for solving problems in different domains and to implement the solutions on a computer system. It mainly emphasizes applying the object-oriented programming (OOP) principles and the need to separate the interface from the implementation. The students will learn the syntax and control structures of the C++ programming language for the implementation.																							
Textbook (or Laboratory Manual for Laboratory Courses)	Tony Gaddis "Starting Out with C++ from Control Structures to Objects" 8 th Edition																							
Reference Material	Paul Deitel, Harvey Deitel "C++ How to Program" 10 th Edition Walter Savitch "Problem Solving with C++" 10 th Edition D. S. Malik "C++ Programming: From Problem Analysis to Program Design" 8 th Edition																							
Course Learning Outcomes	A. Course Learning Outcomes (CLOs) After completion of the course, the students shall be able to: <table border="1"> <thead> <tr> <th>CLOS</th> <th>BT</th> <th>PLO</th> </tr> </thead> <tbody> <tr> <td>1. Demonstrate the basic concepts of OOP</td> <td>3</td> <td>1</td> </tr> <tr> <td>2. Apply OOP concepts (Encapsulation, Inheritance, Polymorphism, Abstraction) to computing problems for the related program</td> <td>3</td> <td>2</td> </tr> <tr> <td>3. Model an algorithmic solution for a given problem using OOP</td> <td>6</td> <td>3</td> </tr> <tr> <td>4. Apply good programming practices</td> <td>3</td> <td>3</td> </tr> </tbody> </table> B. Program Learning Outcomes For each attribute below, indicate whether this attribute is covered in this course or not. Leave the cell blank if the enablement is little or non-existent. <table border="1"> <thead> <tr> <th>PLO 1</th> <th>Computing Knowledge</th> <th>Apply knowledge of mathematics, natural sciences, computing fundamentals, and a computing specialization to the solution of complex computing problems.</th> </tr> </thead> <tbody> <tr> <td></td> <td></td> <td></td> </tr> </tbody> </table>			CLOS	BT	PLO	1. Demonstrate the basic concepts of OOP	3	1	2. Apply OOP concepts (Encapsulation, Inheritance, Polymorphism, Abstraction) to computing problems for the related program	3	2	3. Model an algorithmic solution for a given problem using OOP	6	3	4. Apply good programming practices	3	3	PLO 1	Computing Knowledge	Apply knowledge of mathematics, natural sciences, computing fundamentals, and a computing specialization to the solution of complex computing problems.			
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	PLO 2	Problem Analysis	Identify, formulate, research literature, and analyze complex computing problems, reaching substantiated conclusions using first principles of mathematics, natural sciences, and computing sciences.
	PLO 3	Design/Develop Solutions	Design solutions for complex computing problems and design systems, components, and processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations.
	PLO 4	Investigation & Experimentation	Conduct investigation of complex computing problems using research based knowledge and research based methods
	PLO 5	Modern Tool Usage	Create, select, and apply appropriate techniques, resources and modern computing tools, including prediction and modelling for complex computing problems.
	PLO 6	Society Responsibility	Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal, and cultural issues relevant to context of complex computing problems.
	PLO 7	Environment and Sustainability	Understand and evaluate sustainability and impact of professional computing work in the solution of complex computing problems
	PLO 8	Ethics	Apply ethical principles and commit to professional ethics and responsibilities and norms of computing practice.
	PLO 9	Individual and Team Work	Function effectively as an individual, and as a member or leader in diverse teams and in multi-disciplinary settings.
	PLO 10	Communication	Communicate effectively on complex computing activities with the computing community and with society at large.
	PLO 11	Project Management and Finance	Demonstrate knowledge and understanding of management principles and economic decision making and apply these to one's own work as a member or a team.
	PLO 12	Life Long Learning	Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological changes.

C. Mapping of CLOs on PLOs (CLO: Course Learning Outcome, PLOs: Program Learning Outcomes)											
		PLOs									
		1	2	3	4	5	6	7	8	9	10
CLOs	1	✓									
	2		✓								
	3			✓							
	4			✓							

Topics Covered in the Course, with Number of Lectures on Each Topic (assume 15-week instruction and one-hour lectures)	Topics to be covered:			
	List of Topics	No. of Weeks	Contact Hours	CLO(s)
	Memory Layout, pointers, dynamic memory allocation, stack vs. heap, pointer arithmetic, pointer vs. array, multidimensional pointers, pointer types, char* pointers, alias to pointers (*&)	2	6	1
	Recursion, recursion for patterns and problem solving	1	3	1
	Introduction to object-oriented design, class / struct keyword, member variables, member functions	1	3	2
	Class constructors vs. destructors, copy constructor, overloading constructors, this pointer for function call resolution.	1.5	4.5	2
	constant vs non-constant members, static data members and functions	0.5	1.5	2
	Function overloading, introduction to operator overloading, unary and binary operators overloading, member vs. global functions for operator overloading, stream insertion and extraction operators overloading (friend functions)	1.5	4.5	2
	Identifying classes and defining relationships, introduction to composition (Association & Aggregation), separating interface from implementation	2	6	2,3
	Introduction to inheritance, types of inheritance, function overriding, function overriding vs. overloading, single vs. multiple inheritance	2.5	7.5	2,3
	Introduction to polymorphism, virtual vs. non-virtual functions, abstract vs. concrete classes	1.5	4.5	2,3
	Introduction to templates, template functions and template classes, C++-20 advance concept	1	1.5	1,2,3
	STL	0.5	0.5	1,2,3
	Total	15	45	
Laboratory Projects/Experiments Done in the Course	Yes, there are lab tasks with every lab of three hours.			
Programming Assignments Done in	Yes, there are four programming assignments and a project.			



the Course				
Class Time Spent (in hours)	Theory	Problem Analysis	Solution Design	Social and Ethical Issues
	34	5	5	1
Oral and Written Communications	Every student is required to submit at least __1__ written reports of typically __5__ pages and to make __1__ demonstration of typically __10__ minutes duration.			